GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING
DEPARTMENT OF INFORMATION TECHNOLOGY
INTAKE 41



ESSENTIALS OF ARTIFICIAL INTELLIGENCE
PROJECT PROPOSAL

	Group Details		
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Project Title	AI-Based Herbal Plant Identification and Digital Herbarium System - CeylonHerb		

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1. Project Title

HerbSense - AI-Based Medicinal Plant Identification and Therapeutic Information System
HerbSense is a web-based AI system that identifies medicinal plants from images. Users can upload an image or capture one using a camera. The system predicts the plant using a YOLO11n model and displays the confidence score together with medicinal and botanical information.

2. Problem Statement

Identifying medicinal plants can be difficult for people without botanical knowledge because many plants have similar physical features.

The aim of HerbSense is to provide a simple AI-based solution that can identify medicinal plants from images and provide useful information such as:

- Scientific name
- Medicinal uses
- Active compounds
- Habitat
- Safety precautions

The system is mainly intended for educational and identification purposes.

3. Changes Made After the Proposal

Several improvements have been made during development:

- Developed a complete React-based frontend.
- Added a Node.js and Express backend.
- Integrated the trained YOLO11n model through Roboflow.
- Added image upload and camera capture.
- Added confidence-based prediction results.
- Added alternative plant predictions.
- Added medicinal plant information.
- Improved API security by keeping the Roboflow API key in the backend.
- Added error handling, retry and request validation.

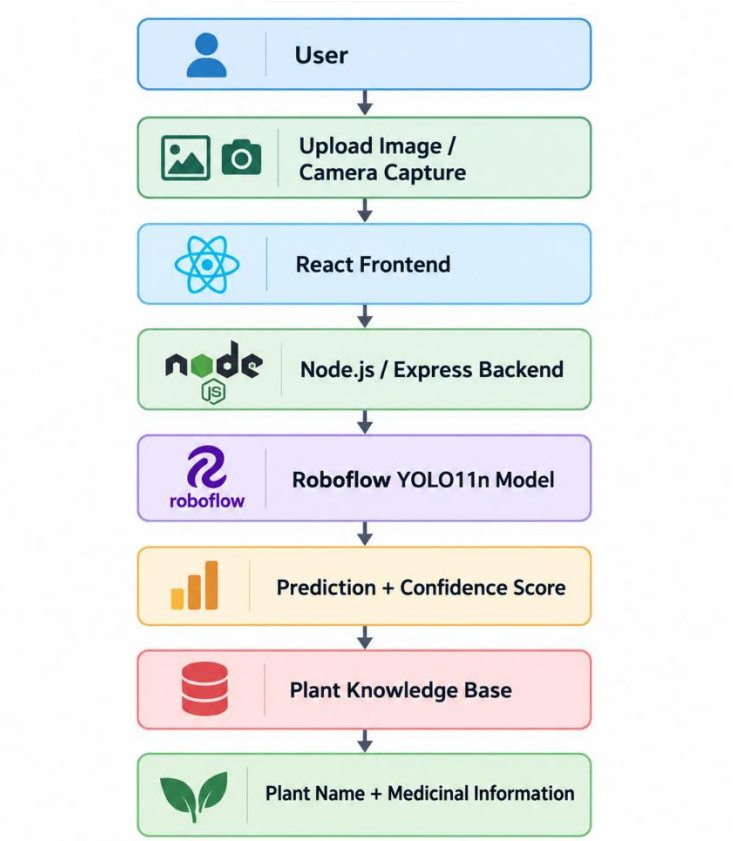
4. Finalized AI Techniques

The main AI technique used is **Deep Learning-based Computer Vision using YOLO11n**.
The model analyses medicinal plant images and returns:

- Predicted plant class
- Confidence score
- Alternative predictions

The system then retrieves matching medicinal information from the plant knowledge base.
The overall approach is:
Deep Learning + Computer Vision + Confidence Rules + Knowledge-Based Information Retrieval

5. System Workflow

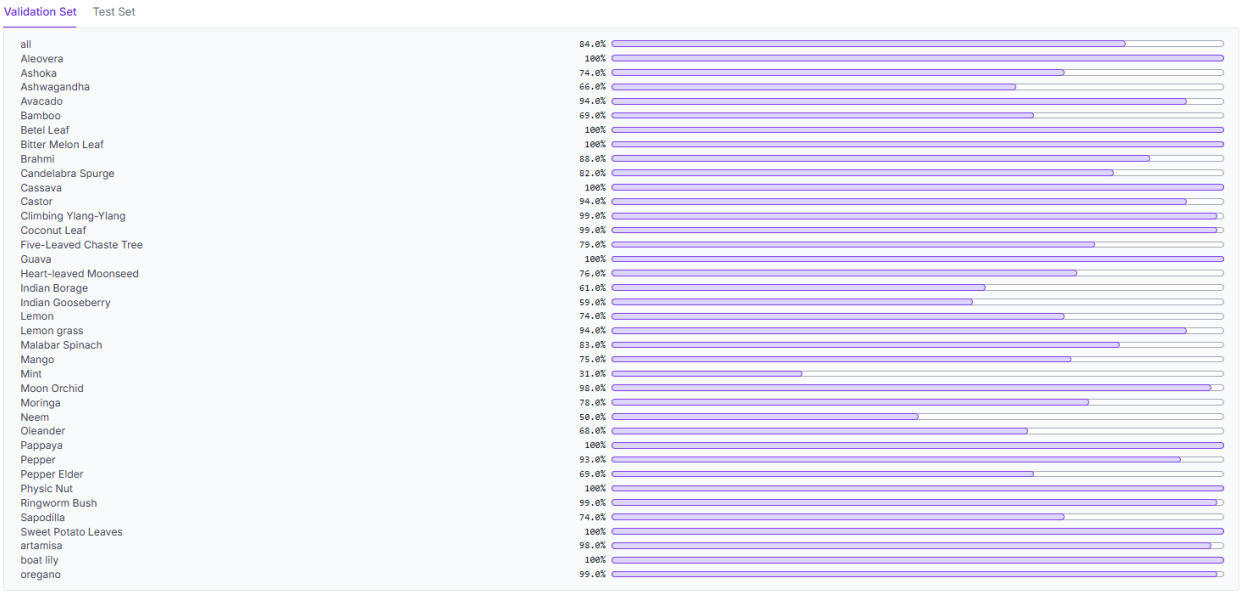


6. Dataset Details

The project uses an image dataset of medicinal plants for model training.

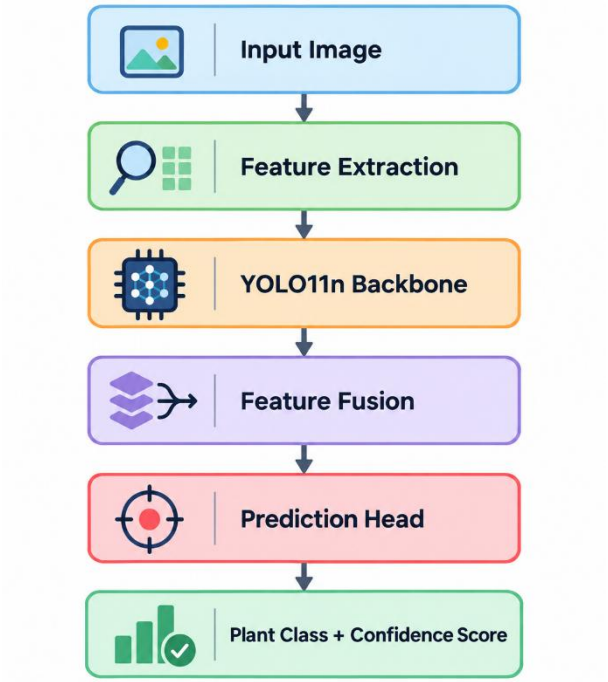
Dataset Platform: Roboflow
Model: YOLO11n
Dataset Type: Medicinal plant images
Dataset Information

- Total images: **5915**
- Number of classes: **39**
- Training images: **4488**
- Validation images: **845**
- Testing images: **582**
- Dataset version: **V1**



7. Neural Network Structure

The YOLO11n model follows the basic structure:



The backbone extracts important features from the plant image, while the prediction head produces the final plant identification.

8. Rules Used in the System

The system uses confidence-based rules.
IF confidence >= 75%
THEN High Confidence

IF confidence >= 45% AND confidence < 75%
THEN Moderate Match

IF confidence < 45%
THEN Low Confidence
The plant with the highest confidence value is selected as the main prediction.

9. Partial Implementation Completed

The following functions have already been implemented:

- Image upload
- Camera capture
- Image preview
- AI plant identification
- Confidence score display
- Alternative predictions
- Medicinal plant information
- React frontend
- Express backend
- Roboflow integration
- Error handling
- API security
- Responsive interface

10. Screenshots / Sample Outputs

Add the following screenshots to the report:

Figure 1: HerbSense Home Page

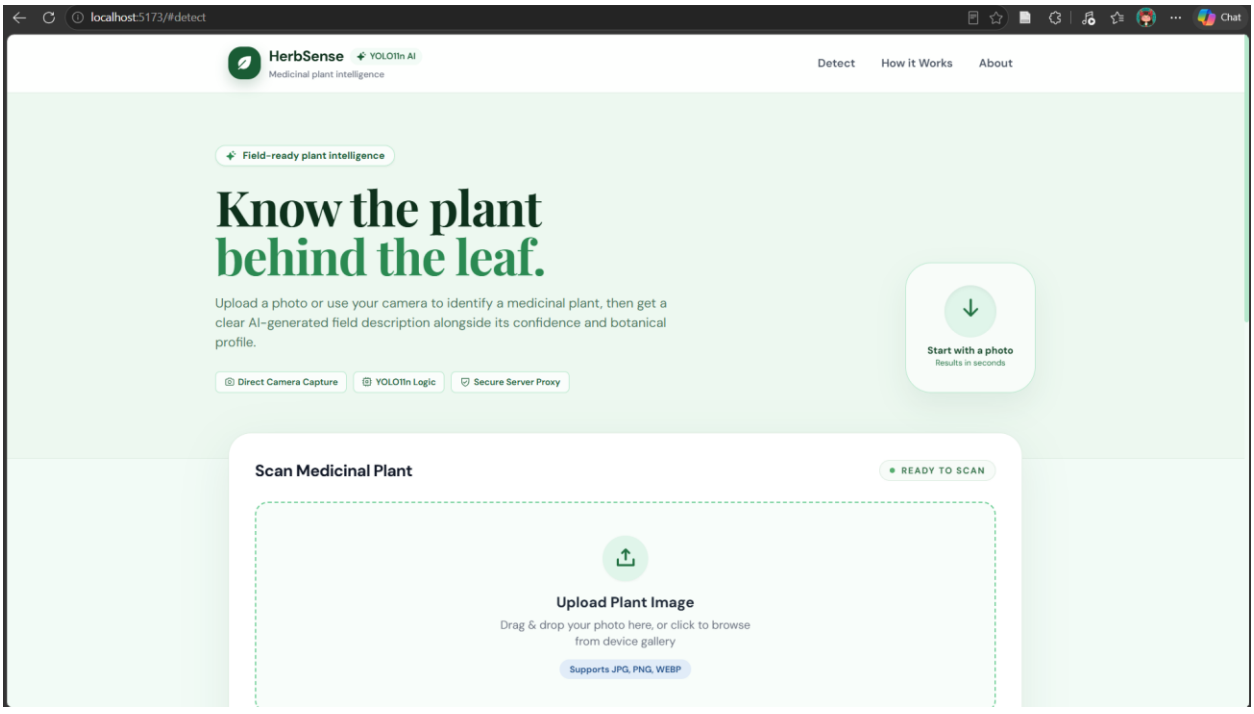


Figure 2: Image Upload / Camera Interface

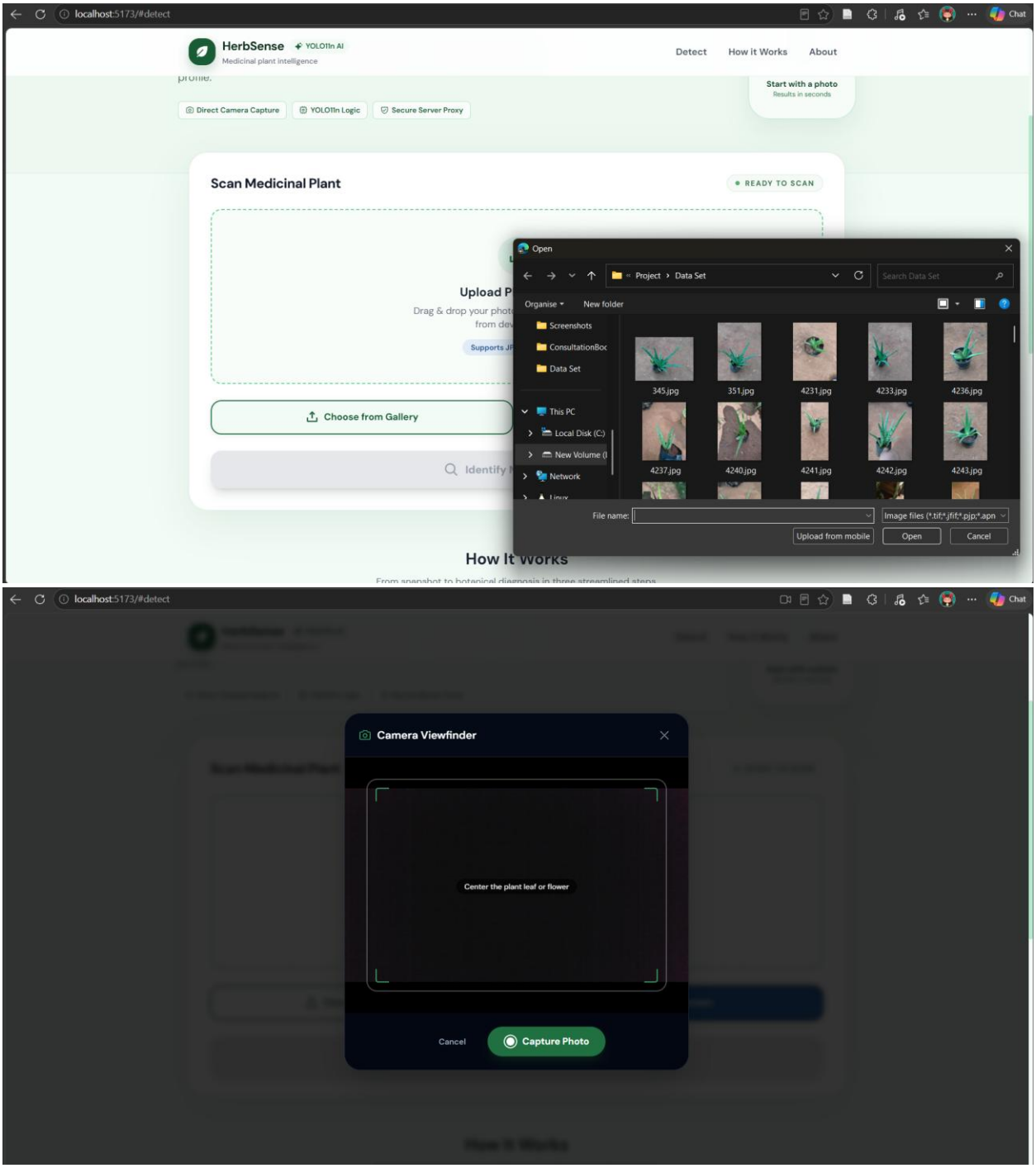


Figure 3: Selected Plant Image

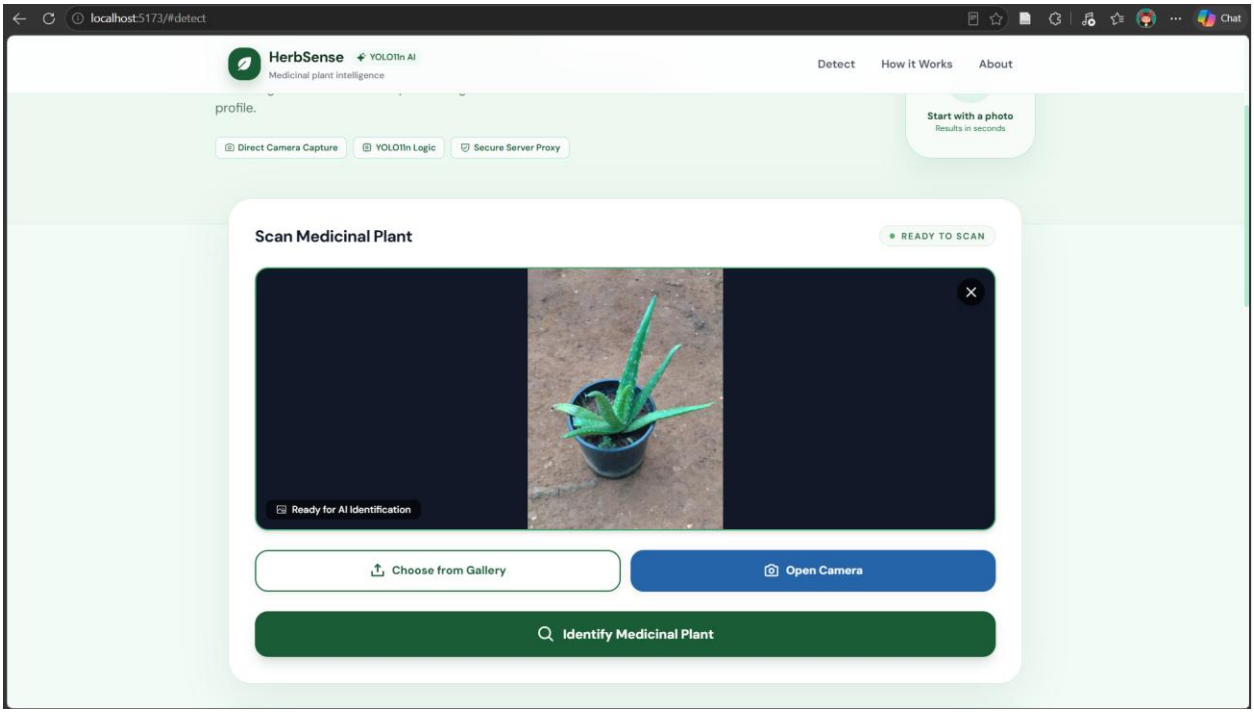
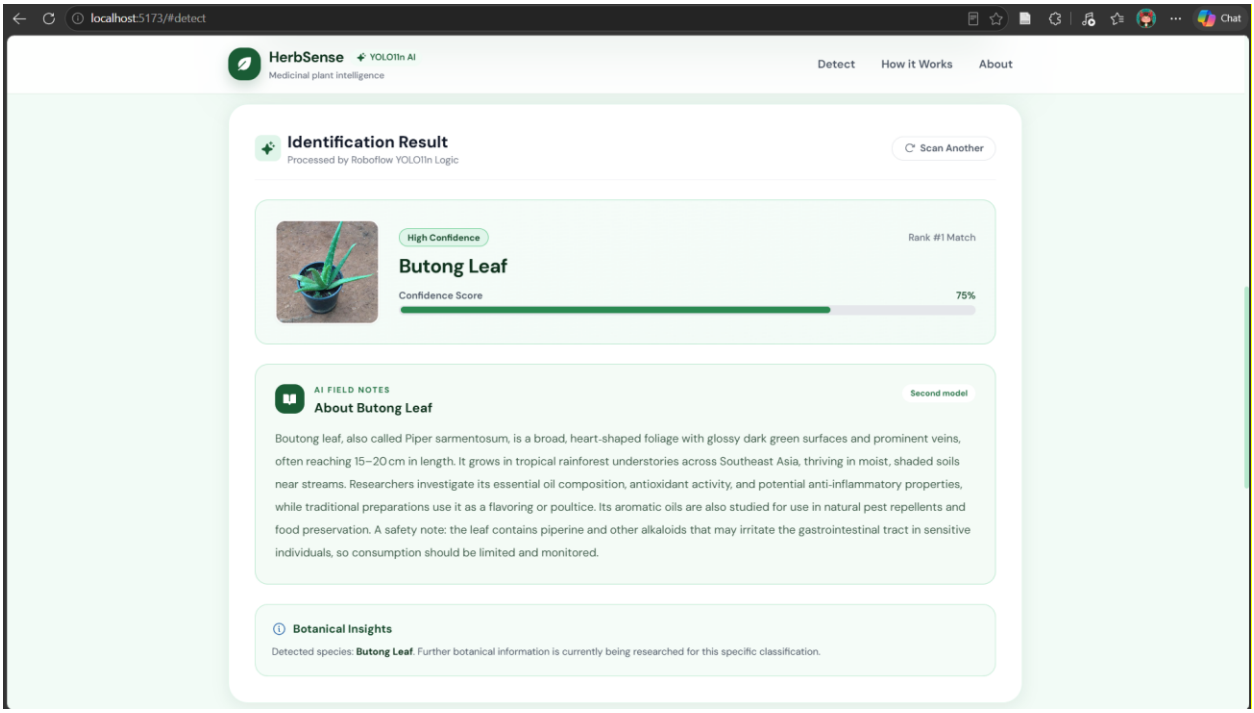
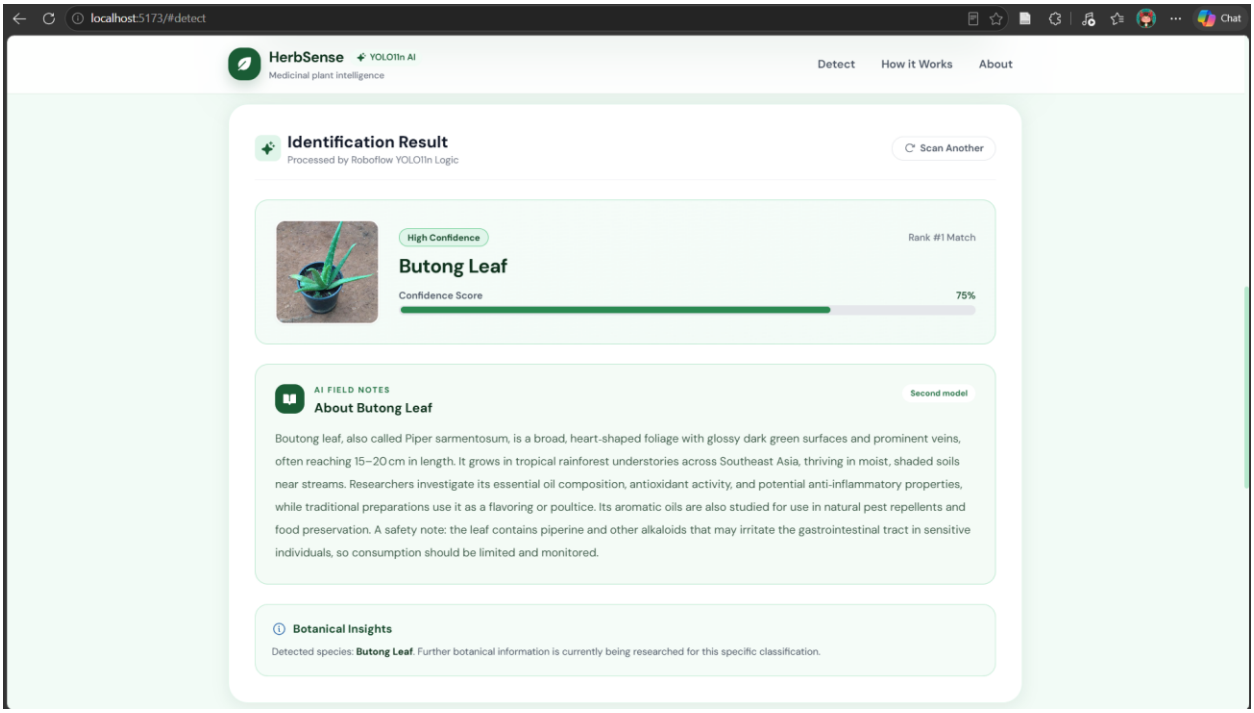


Figure 4: Plant Identification Result



We will correct the name and details of the plant in future developments.

Figure 5: Confidence Score and Alternative Predictions



Alternative Predictions will develop in future developments.

Figure 6: Medicinal Plant Information

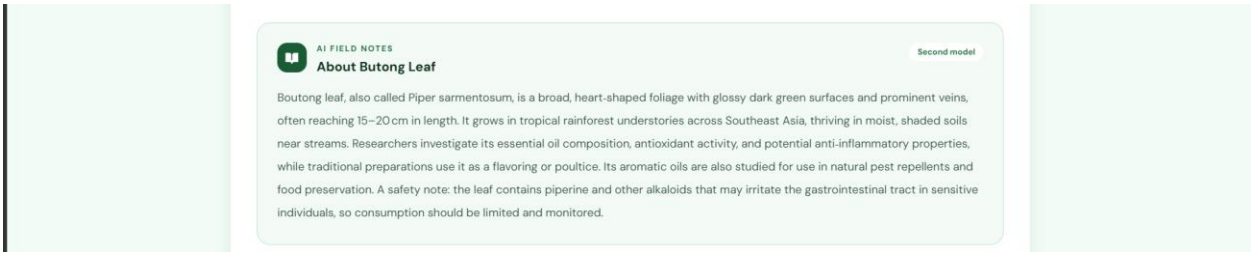
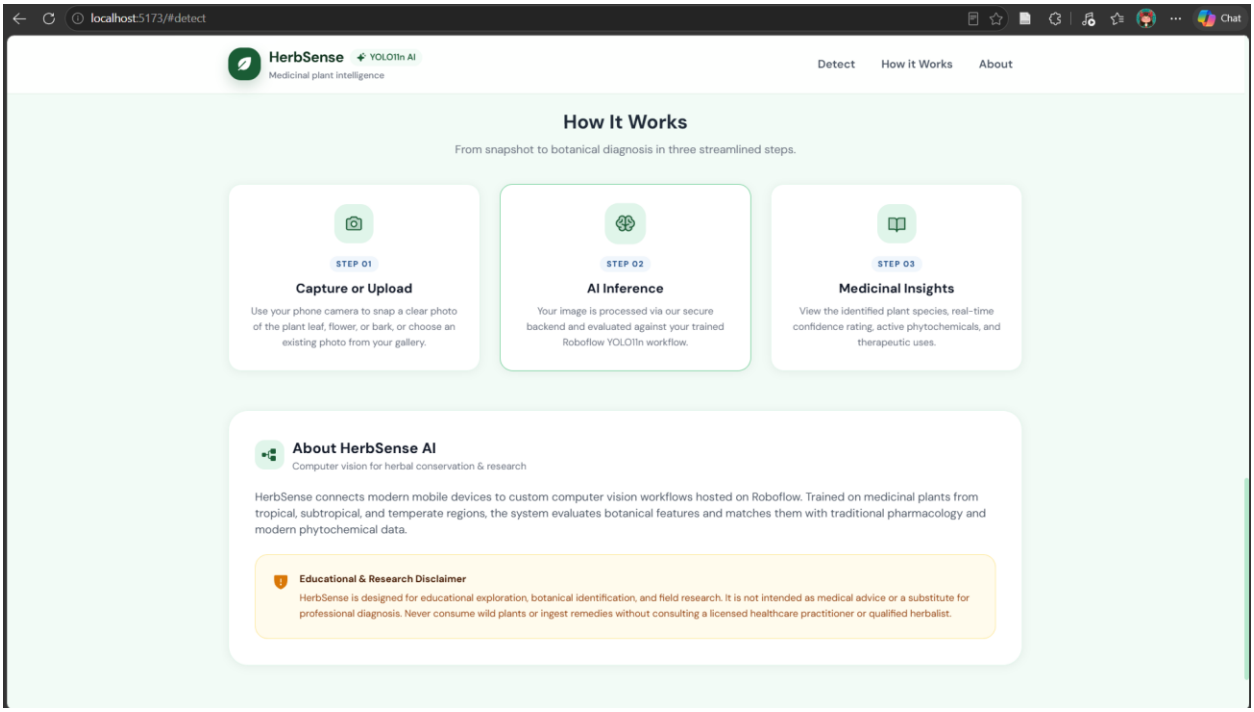


Figure 7: How It Works



11. Problems Faced During Development

Several challenges were encountered during the development of the HerbSense system. These include:

- Integrating the Roboflow AI model with the web application.
- Securing the Roboflow API key to prevent unauthorized access.
- Handling different prediction response formats received from the AI model.
- Managing network failures and API communication errors.
- Processing large images captured through the camera.
- Managing AI model API usage limitations.
- Adding accurate medicinal information for each identified plant class.
- Improving the accuracy of plant names and details displayed by the system.

These challenges were minimized through backend API integration, input validation, error handling mechanisms, retry methods, and continuous improvement of the plant information database.

12. Remaining Work

Before the final submission, the following tasks must be completed:

1. Final model testing using unseen images and different environmental conditions.
2. Add dataset statistics and complete model evaluation results (precision, recall, and mAP).
3. Verify and improve medicinal plant information and displayed plant details.
4. Implement alternative prediction suggestions for uncertain results.
5. Complete frontend and backend testing and improve mobile responsiveness.
6. Deploy the final system and capture final outputs.
7. Complete documentation and presentation preparation.

13. Updated Member Contributions

Member	Contribution
KMGPD Kariyawasam	Dataset collection, image labelling, dataset preprocessing, and model training
PV Kavinda	YOLO11n integration, backend development, API connection, and error handling
WGISK Wedage	React frontend development, camera function, image upload, and results display
KHTU Jayasekara	Medicinal plant information management, system testing, documentation, and GitHub management

14. Current Project Status

The core HerbSense system has been partially completed and successfully integrated with the AI prediction model.

The current workflow is:
Plant Image → AI Model → Plant Prediction → Confidence Score → Medicinal Information

The main system functionalities, including image input, AI-based plant identification, and displaying medicinal information, have been implemented.
The remaining improvements include final model evaluation, improving plant information accuracy, adding alternative prediction options, completing system testing, deployment, and preparing the final project submission.